

Week 2 — PDET Municipality Boundaries

Dataset Integration

Course: Database Administration **Deliverable:** Week 2 — PDET Municipality Boundaries **Date:** 2026-05-11 **Repository:** `project/`

1. Scope

This deliverable integrates the official set of 170 PDET municipalities (*Programas de Desarrollo con Enfoque Territorial*, Decreto 893 de 2017) into a MongoDB collection with spatial indexing, ready for the building footprint join in Week 3.

The deliverable is evaluated against four checkpoints:

- 1. Data Acquisition & Verification
- 2. Data Integrity & Format
- 3. NoSQL Spatial Integration
- 4. Documentation of Process

Each section below documents the evidence for one checkpoint.

2. Infrastructure

The project runs in Docker. MongoDB 7 and Mongo Express are orchestrated through `docker-compose.yml`. Bringing the stack up takes a single command.

☐	Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
☐	project	-	-	-	0.43%	13 hours ago	   
☐	upme-mongo-express	e6335f315b94	mongo-express:1.0.2	8081:8081 	0%	13 hours ago	   
☐	upme-mongo	1d2775bb5dd4	mongo:7	27017:27017 	0.43%	13 hours ago	   

The `upme-mongo` service exposes MongoDB on port 27017 and `upme-mongo-express` exposes the browser UI on port 8081. Both containers are healthy; the database is reachable.

3. Checkpoint 1 — Data Acquisition & Verification

Two authoritative sources were combined:

Dataset	Provider	Role
MGN 2025 (Marco Geoestadístico Nacional)	DANE	Polygon geometry for all 1,122 Colombian municipalities, EPSG:4326
MunicipiosPDET.xlsx	ART (Agencia de Renovación del Territorio)	Legal list of 170 PDET municipalities per Decreto 893/2017

Both sources are documented in `data/raw/SOURCES.md` (URLs, download dates, licenses, and a reproduction recipe).

The ART list was cross-joined against the MGN GeoJSON by DIVIPOLA code. **Result: 170 / 170 codes matched. Zero orphans.** This eliminates any risk of unmatched names or stale municipal codes.

4. Checkpoint 2 — Data Integrity & Format

`scripts/validate_pdet.py` runs eleven automated checks and writes the result to `docs/week2-validation.md` on every execution. The audit log re-generates each run, so it is always synchronized with the data on disk.

```
PS C:\Users\aleja\Desktop\universidad\project> python scripts/validate_pdet.py
[*] Loading data\raw\municipios_colombia.geojson ...
    1122 features, CRS=EPSG:4326
[OK] CRS is EPSG:4326: True
[OK] Source columns present: missing=[]
[*] PDET reference list: 170 codes (170 expected)
[OK] PDET codes found in MGN: 170/170
[OK] No duplicate DIVIPOLA in filtered set: True
[OK] Geometry types: {'MultiPolygon': 170}
[OK] Invalid geometries: 0
[OK] Empty geometries: 0
[OK] Bounds within Colombia BBOX: (-79.010,-0.706,-69.995,11.349)
[*] Reprojecting to EPSG:9377 for equal-area computation ...
    Total PDET area: 389,182.0 km^2 (sanity: 300,000-500,000)
[OK] Total area within expected range: True
    Smallest muni: 83.67 km^2
    Largest muni: 42,233.53 km^2
[OK] PDET subregions: 16 (16 expected)
[OK] Wrote data\processed\pdet_municipios.geojson (65.2 MB, 170 features)
[OK] Wrote docs/week2-validation.md
```

Checks performed:

1. **CRS** — input is EPSG:4326.
2. **Required source columns** present (`mpio_cdpmp` , `mpio_cnmb` , `dpto_cnmb` , `dpto_ccdgo` , `mpio_ccdgo`).
3. **170 / 170 PDET DIVIPOLA codes** present in the MGN dataset.
4. **No duplicate DIVIPOLA** in the filtered subset.
5. **Geometry types** restricted to Polygon / MultiPolygon (result: 170 MultiPolygons).
6. **Geometry validity** (`shapely.is_valid`) — 0 invalid geometries. `make_valid` is applied defensively if any are found.

7. **No empty geometries.**
8. **Bounds inside Colombia's bounding box.**
9. **Total area in sanity range** — computed 389,182 km² ($\approx 34\%$ of national territory, consistent with ART's published figures).
10. **16 PDET subregiones** present after merge (matches Decreto 893/2017).
11. **Cleaned output file written** — `data/processed/pdet_municipios.geojson`.

Coordinate-reference decisions

Stored geometry remains in **EPSG:4326** because MongoDB's `2dsphere` index requires it. Area cannot be measured in degrees, so polygons are reprojected to **EPSG:9377 (MAGNA-SIRGAS Origen Nacional)** — Colombia's official equal-area CRS — and the resulting `area_sqkm` is stored as a plain number on each document. Queries therefore never need to reproject.

5. Checkpoint 3 — NoSQL Spatial Integration

`scripts/load_municipalities.py` upserts the 170 cleaned polygons into the `upme.municipalities` collection, keyed on DIVIPOLA. The load is idempotent: re-running replaces documents in place.

Collection state

The Mongo Express UI confirms 170 documents in the collection:

Mongo Express Database: upme Collection: municipalities

Viewing Collection: municipalities

[New Document](#) [New Index](#)

[Simple](#) [Advanced](#)

Key Value String [Find](#)

Delete all 170 documents retrieved

1 2 3 4 5 > >>

_id	divipola	name	department	department_code	is_pdet	area_sqkm	source	loaded_at	geometry
69f66f9c5a598acca412f129	52835	SAN ANDRÉS DE TUMACO	NARIÑO	52	true	3617.458058	DANE MGN2025 + ART PDET (Decreto 893/2017)	2026-05-11T04:38:27+00:00	*** LARGE PROPERTY *** ~798 KB Preview: {"type": "MultiPolygon"} Click to fetch this property
6a015d14b5bb5487eb70530e	05031	AMALFI	ANTIOQUIA	05	true	1208.346188	DANE MGN2025 + ART PDET (Decreto 893/2017)	2026-05-11T04:38:27+00:00	*** LARGE PROPERTY *** ~609 KB Preview: {"type": "MultiPolygon"} Click to fetch this property
6a015d14b5bb5487eb70530e	05040	ANORÍ	ANTIOQUIA	05	true	1412.956957	DANE MGN2025 + ART PDET (Decreto 893/2017)	2026-05-11T04:38:27+00:00	*** LARGE PROPERTY *** ~609 KB Preview: {"type": "MultiPolygon"} Click to fetch this property

Document shape

Each document follows the `$jsonSchema` validator installed by `scripts/init.js`. Below is a single document expanded in Mongo Express, showing the canonical field set (`divipola` , `name` , `department` , `is_pdet` , `area_sqkm` , `subregion_pdet` , `source` , `loaded_at` , `geometry`).

The screenshot shows the Mongo Express web interface. At the top, it says 'Mongo Express Database: upme Collection: municipalities'. Below this is a header 'Viewing Collection: municipalities'. There are buttons for 'New Document' and 'New Index'. A search bar contains 'divipola' and '52835', with a 'Find' button. A red banner indicates 'Delete all 1 documents retrieved'. Below this is a table with columns: `divipola`, `name`, `department`, `department_code`, `is_pdet`, `area_sqkm`, `source`, `loaded_at`, `geometry`, and `subregion_pdet`. The table contains one document with the following values: `divipola`: 52835, `name`: SAN ANDRÉS DE TUMACO, `department`: NARIÑO, `department_code`: 52, `is_pdet`: true, `area_sqkm`: 3617.458058, `source`: DANE MGN2025 + ART PDET (Decreto 893/2017), `loaded_at`: 2026-05-11T04:38:27+00:00, `geometry`: a GeoJSON MultiPolygon object, and `subregion_pdet`: PACÍFICO Y FRONTERA NARIÑENSE. Below the table is a 'Rename Collection' button.

divipola	name	department	department_code	is_pdet	area_sqkm	source	loaded_at	geometry	subregion_pdet
52835	SAN ANDRÉS DE TUMACO	NARIÑO	52	true	3617.458058	DANE MGN2025 + ART PDET (Decreto 893/2017)	2026-05-11T04:38:27+00:00	{ "type": "MultiPolygon", "coordinates": [[[[[[[[[[[[[]]]]]]]]]]]] }	PACÍFICO Y FRONTERA NARIÑENSE

Indexes and spatial query proof

Three indexes are provisioned at initialization time:

- `geometry_2dsphere` — spherical spatial index for `$geoWithin` / `$geoIntersects`.
- `divipola_1` (*unique*) — upsert key and lookup index.
- `is_pdet_1` — selective filter for downstream queries.

The screenshots below show the indexes and two `$geoIntersects` queries that together prove both the index and the PDET scoping are correct.

Indexes listed in mongosh:

```
PS C:\Users\aleja\Desktop\universidad\project> docker compose exec mongo mongosh -u root -p root --authenticationDatabase admin --quiet upme --eval "db.municipalities.getIndexes().map(i => i.name)"
[ ' _id_', 'geometry_2dsphere', 'divipola_1', 'is_pdet_1' ]
```

`$geoIntersects` from a point inside Tumaco (DIVIPOLA 52835 — a PDET municipality):

```
upme> db.municipalities.countDocuments({ geometry: { $geoIntersects: { $geometry: { type: "Point", coordinates: [-78.69039, 1.61122] } } } })
1
```

Result: 1 — the query correctly returns San Andrés de Tumaco.

`$geoIntersects` from a point inside Bogotá D.C. (NOT a PDET municipality):

```
upme> db.municipalities.countDocuments({ geometry: { $geoIntersects: { $geometry: { type: "Point", coordinates: [-74.07, 4.60] } } } })
0
```

Result: **0** — the collection correctly excludes Bogotá, proving the PDET-only scoping is honored end-to-end.

The Bogotá miss is as load-bearing as the Tumaco hit. If the loader had inadvertently inserted all 1,122 MGN municipalities, the same query would return Bogotá D.C. The zero result confirms scope correctness.

6. Checkpoint 4 — Documentation of Process

Three documents describe how this deliverable was built and how to re-run it:

- `data/raw/SOURCES.md` — provenance, licenses, download dates, reproduction steps for both raw sources.
- `docs/week2-validation.md` — auto-generated audit log of the 11 integrity checks. Regenerates on every `validate_pdet.py` run.
- `docs/week2-pdet-loading.md` — full methodology write-up, including the rationale for CRS choices, the spatial-join-at-load pattern, and caveats.

The pipeline reproduces from a clean checkout in five commands:

```
docker compose up -d
docker compose exec mongo mongosh -u root -p root \
  --authenticationDatabase admin /scripts/init.js
# (place data/raw/municipios_colombia.geojson per SOURCES.md)
python scripts/validate_pdet.py
python scripts/load_municipalities.py
```

`load_municipalities.py` finishes with the same spatial sanity queries shown in Section 5, so a successful run always demonstrates the 2dsphere index is live.

7. Summary

Checkpoint	Evidence	Status
Data Acquisition & Verification	Two sources, 170/170 join, <code>SOURCES.md</code>	Complete
Data Integrity & Format	11 / 11 automated checks pass; <code>week2-validation.md</code>	Complete
NoSQL Spatial Integration	170 docs, 4 indexes, <code>\$geoIntersects</code> proves Tumaco hit + Bogotá miss	Complete
Documentation of Process	<code>SOURCES.md</code> , <code>week2-validation.md</code> , <code>week2-pdet-loading.md</code> , 5-command reproduce path	Complete

The PDET municipality collection is loaded, indexed, and verified. It exposes the contract that Week 3's building loader will consume: every building's containing PDET municipality can be resolved with a single indexed `$geoIntersects` lookup against `upme.municipalities` .

Appendix A — Top 10 PDET municipalities by area

DIVIPOLA	Name	Department	Area (km²)
18756	SOLANO	CAQUETÁ	42,233.5
18753	SAN VICENTE DEL CAGUÁN	CAQUETÁ	20,170.1
95001	SAN JOSÉ DEL GUAVIARE	GUAVIARE	16,712.4
95015	CALAMAR	GUAVIARE	13,510.3
95200	MIRAFLORES	GUAVIARE	12,774.7
18150	CARTAGENA DEL CHAIRÁ	CAQUETÁ	12,771.4
95025	EL RETORNO	GUAVIARE	12,405.9
50325	MAPIRIPÁN	META	11,917.5
86573	PUERTO LEGUÍZAMO	PUTUMAYO	10,900.4
50350	LA MACARENA	META	8,108.5

Appendix B — PDET subregion breakdown

Subregión PDET	Munis	Área (km ²)
MACARENA - GUAVIARE	12	96,169
CUENCA DEL CAGUÁN Y PIEDEMONTE CAQUETEÑO	17	93,298
CHOCÓ	14	29,548
PUTUMAYO	9	25,063
SIERRA NEVADA - PERIJÁ	15	20,415
BAJO CAUCA Y NORDESTE ANTIOQUEÑO	13	17,615
PACÍFICO Y FRONTERA NARIÑENSE	11	17,119
PACÍFICO MEDIO	4	14,302
ALTO PATÍA Y NORTE DEL CAUCA	24	13,178
ARAUCA	4	10,540
SUR DE BOLÍVAR	7	10,332
SUR DE CÓRDOBA	5	9,581
URABÁ ANTIOQUEÑO	8	9,459
CATATUMBO	8	9,238
SUR DEL TOLIMA	4	6,919
MONTES DE MARÍA	15	6,407
TOTAL	170	389,182